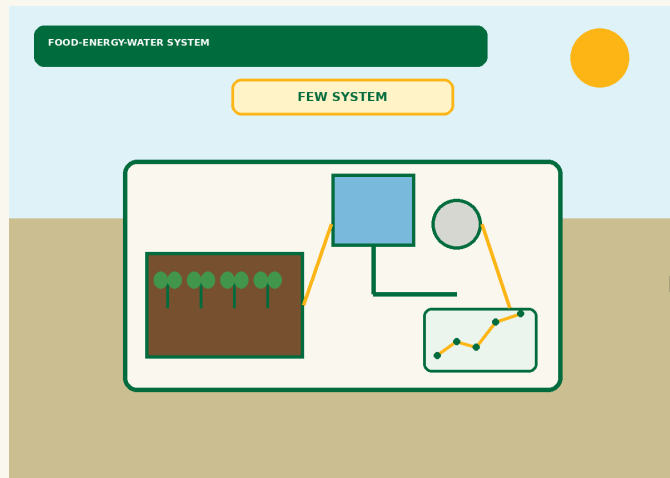




FEW SYSTEM

FOOD-ENERGY-WATER SYSTEM

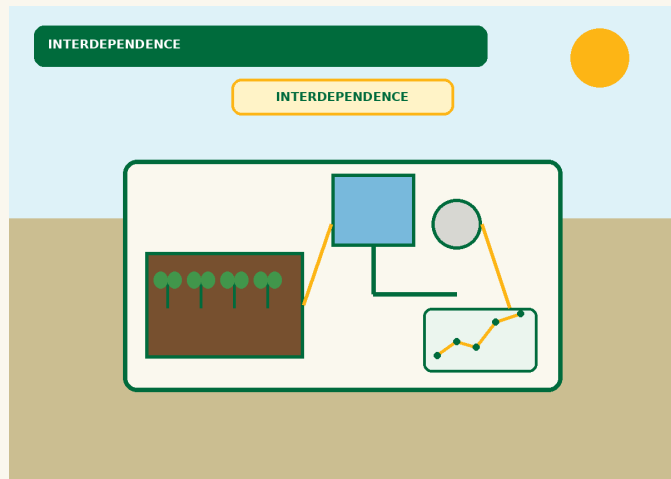
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A food-energy-water system represents interacting production, energy, and water subsystems whose decisions and constraints can propagate across the whole system.

ANALYSIS / DESIGN TASK

Draw the principal subsystem connections.



INTERDEPENDENCE

INTERDEPENDENCE

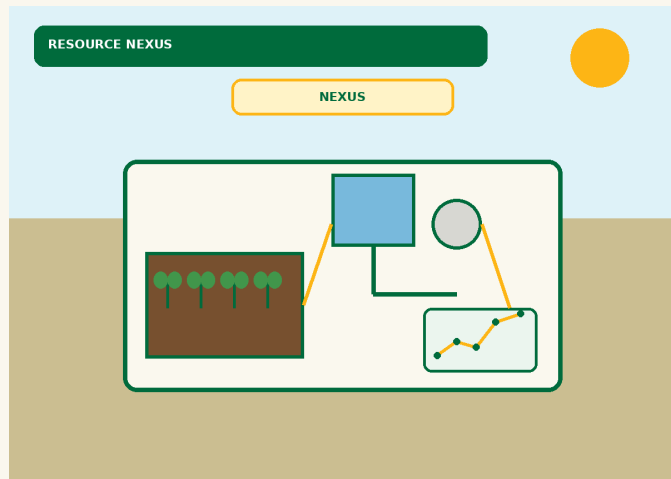
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Interdependence means the condition or performance of one subsystem depends partly on another subsystem.

ANALYSIS / DESIGN TASK

Identify two directional dependencies.



NEXUS

RESOURCE NEXUS

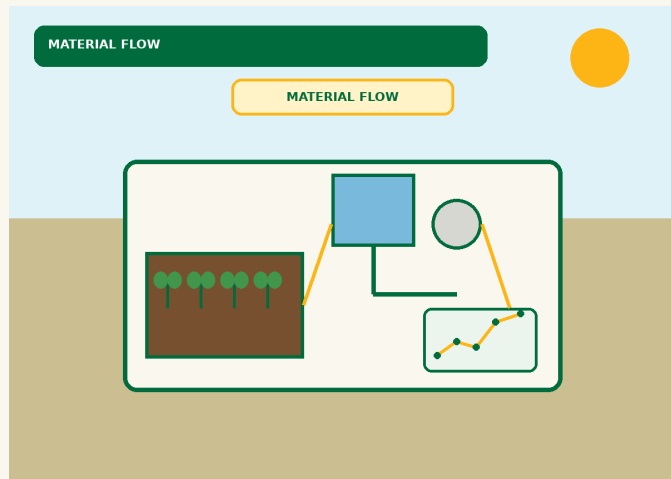
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A resource nexus perspective evaluates linked resource flows and decisions rather than treating food, energy, and water independently.

ANALYSIS / DESIGN TASK

Trace one cross-resource pathway.



MATERIAL FLOW

MATERIAL FLOW

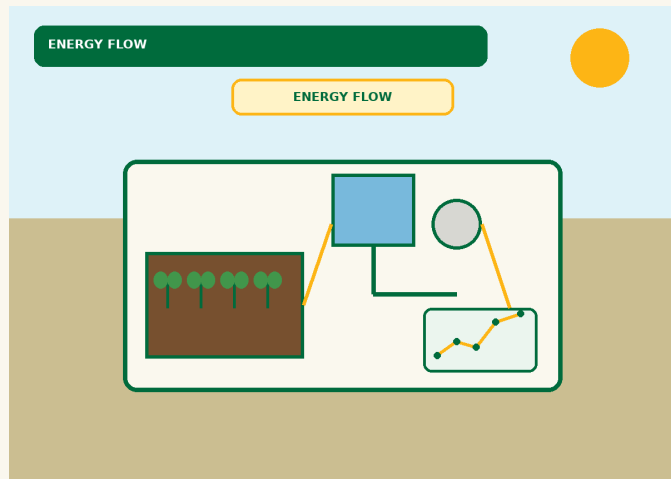
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Material-flow analysis tracks physical quantities entering, moving through, accumulating in, and leaving a defined system.

ANALYSIS / DESIGN TASK

Construct a simple water or nutrient flow table.



ENERGY FLOW

ENERGY FLOW

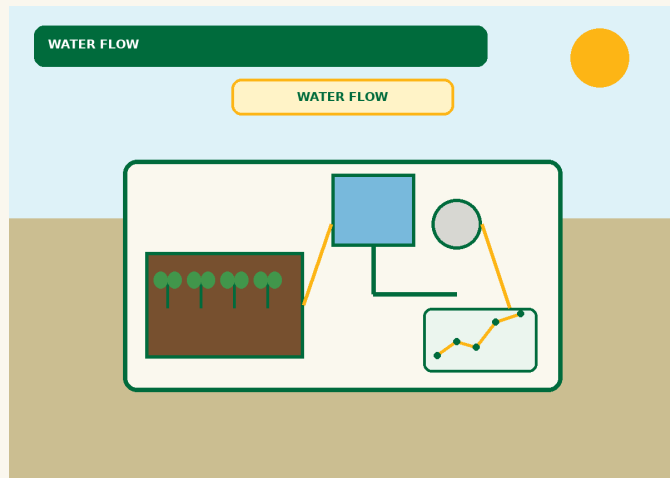
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Energy-flow analysis tracks energy supply, conversion, storage, demand, losses, and useful services within the system.

ANALYSIS / DESIGN TASK

Map energy used for pumping and other loads.



WATER FLOW

WATER FLOW

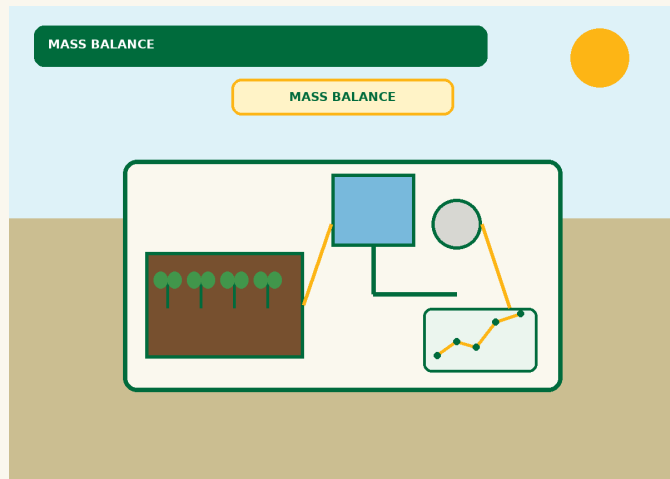
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Water-flow analysis tracks source, storage, conveyance, irrigation, soil storage, crop use, drainage, and losses.

ANALYSIS / DESIGN TASK

Create a system water-flow diagram.



MASS BALANCE

MASS BALANCE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

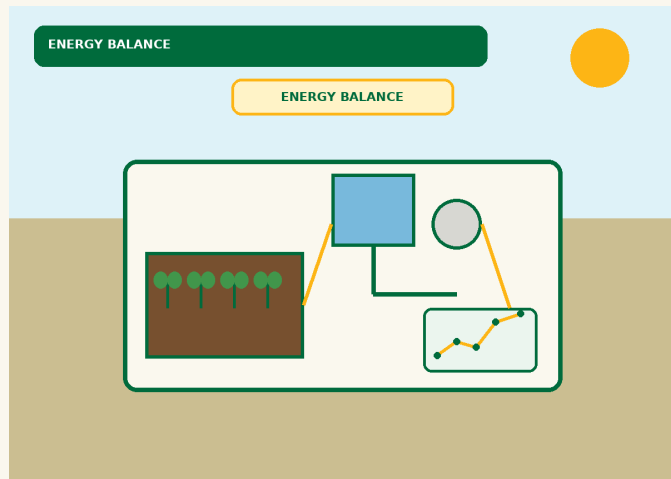
TECHNICAL CONCEPT

A mass balance applies conservation: accumulation equals inputs minus outputs plus or minus internal generation or consumption where relevant.

ANALYSIS / DESIGN TASK

Complete a simplified tank or nutrient balance.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



ENERGY BALANCE

ENERGY BALANCE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

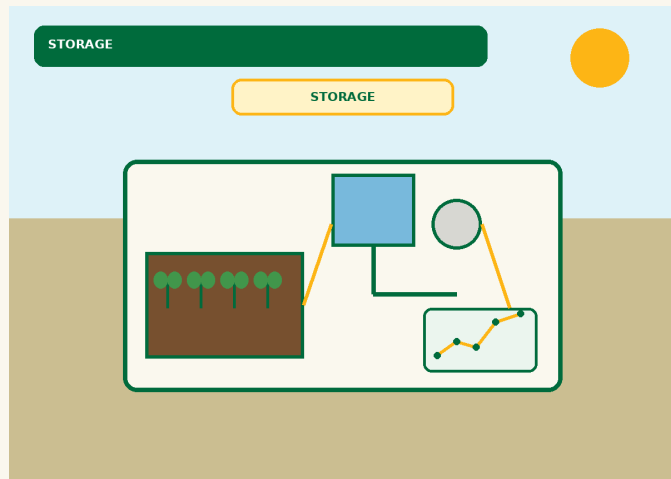
TECHNICAL CONCEPT

An energy balance accounts for energy entering, stored, converted, used, and lost within a defined boundary.

ANALYSIS / DESIGN TASK

Complete a simplified daily energy balance.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



STORAGE

STORAGE

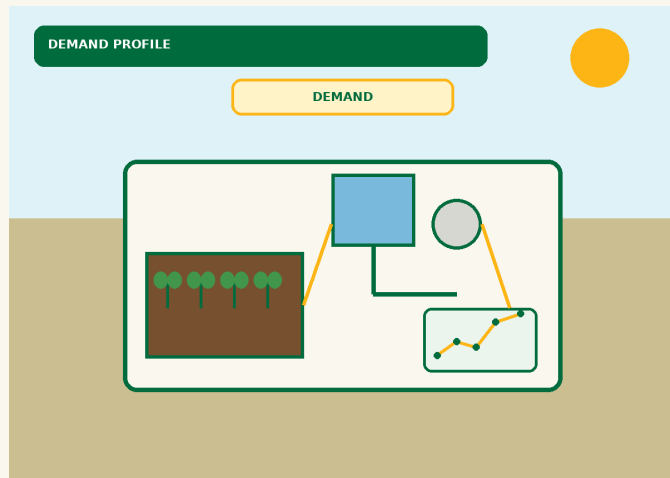
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Storage decouples the timing of resource supply and demand, such as water tanks, batteries, soil water, or harvested products.

ANALYSIS / DESIGN TASK

Compare two storage functions.



DEMAND

DEMAND PROFILE

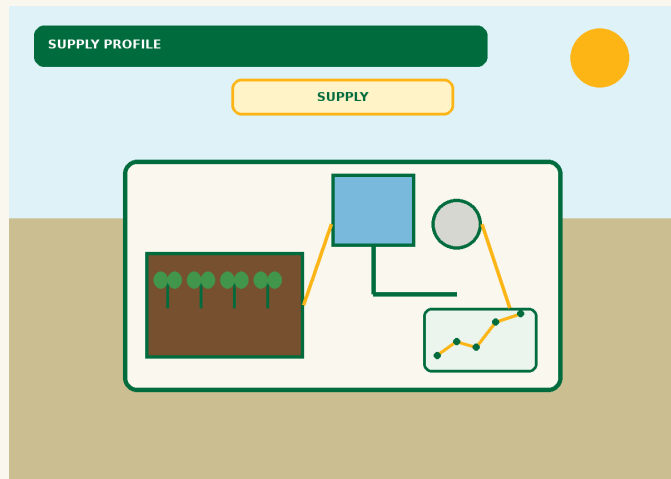
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A demand profile describes how resource or service requirements vary through time.

ANALYSIS / DESIGN TASK

Plot a daily irrigation or electrical demand profile.



SUPPLY

SUPPLY PROFILE

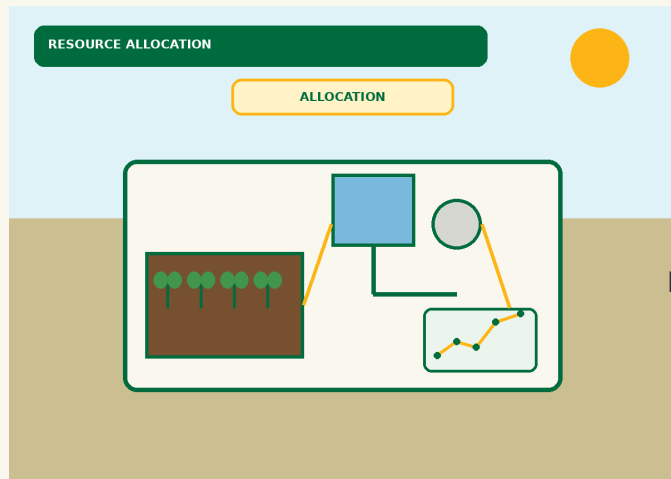
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A supply profile describes how available resource supply changes through time and may not coincide with demand.

ANALYSIS / DESIGN TASK

Compare supply and demand timing.



ALLOCATION

RESOURCE ALLOCATION

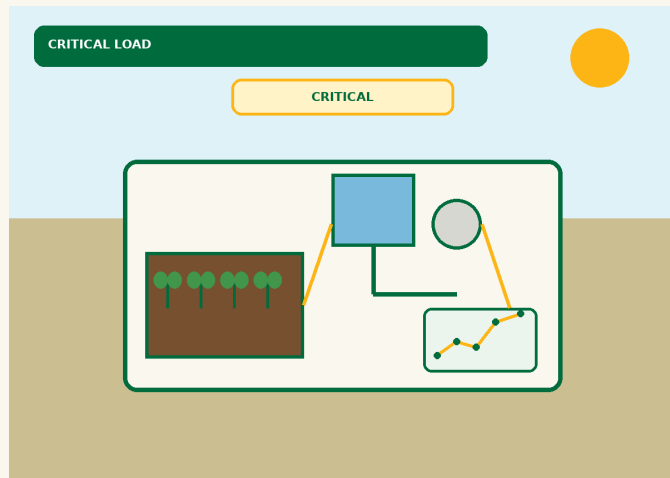
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Resource allocation assigns limited water, energy, labor, or storage capacity among competing uses according to objectives and priorities.

ANALYSIS / DESIGN TASK

Solve a constrained allocation scenario.



CRITICAL

CRITICAL LOAD

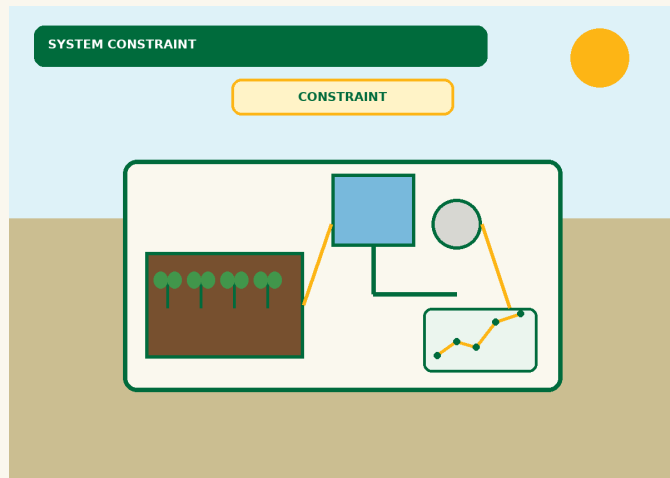
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A critical load or service has high priority because interruption creates unacceptable production, safety, or system consequences.

ANALYSIS / DESIGN TASK

Classify loads by criticality.



CONSTRAINT

SYSTEM CONSTRAINT

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

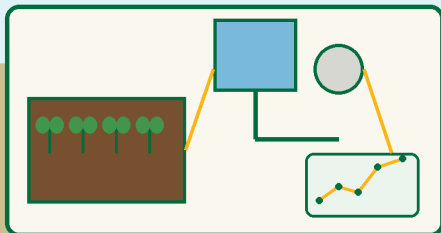
Integrated constraints include limited water, energy, storage, pumping capacity, time, budget, or infrastructure.

ANALYSIS / DESIGN TASK

Identify the binding constraint in a scenario.

CASCADE EFFECT

CASCADE



CASCADE

CASCADE EFFECT

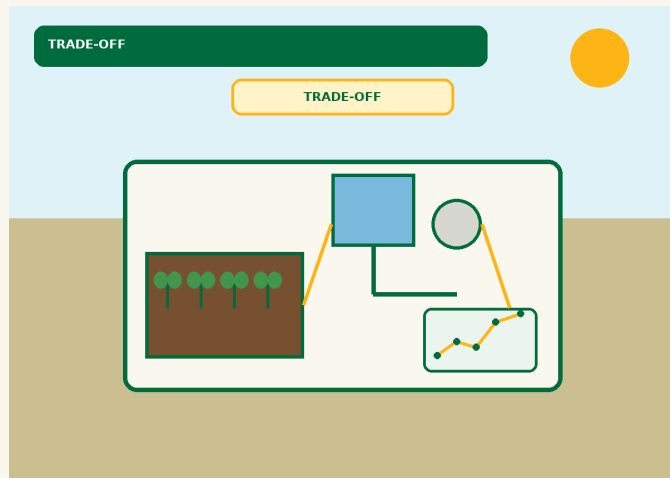
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A cascade occurs when a change or failure in one subsystem propagates through dependencies to affect other functions.

ANALYSIS / DESIGN TASK

Trace a pump-failure cascade.



TRADE-OFF

TRADE-OFF

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

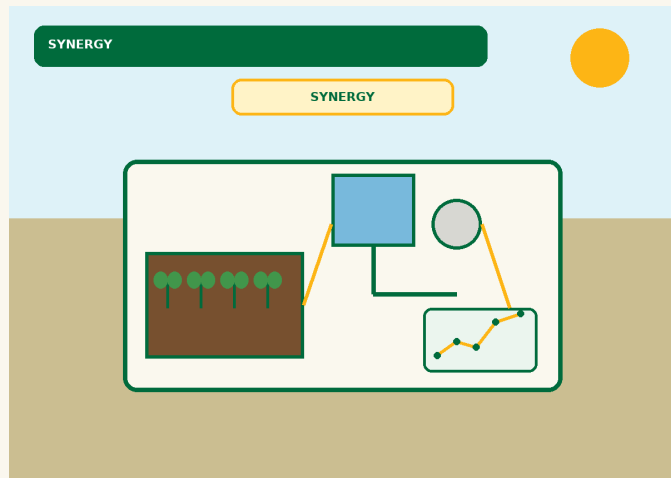
TECHNICAL CONCEPT

Integrated decisions can create trade-offs among production, water use, energy demand, cost, reliability, and other objectives.

ANALYSIS / DESIGN TASK

Compare two management strategies.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



SYNERGY

SYNERGY

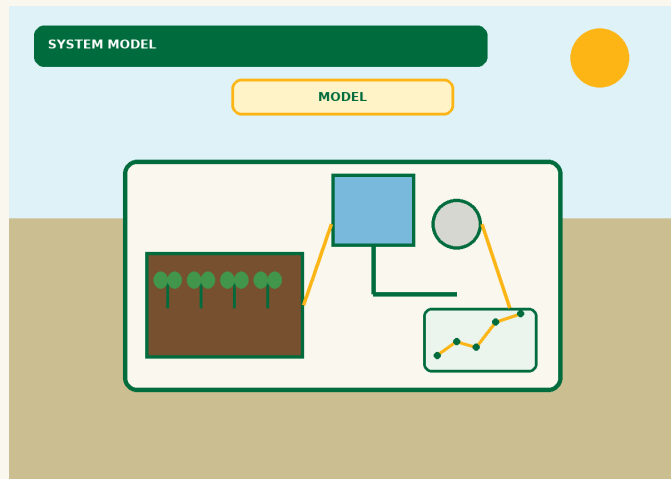
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A synergy occurs when coordinated management improves more than one subsystem objective without an equivalent penalty elsewhere.

ANALYSIS / DESIGN TASK

Identify a plausible coordinated benefit.



MODEL

SYSTEM MODEL

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A system model is a simplified representation of components, states, flows, relationships, and rules used to analyze behavior.

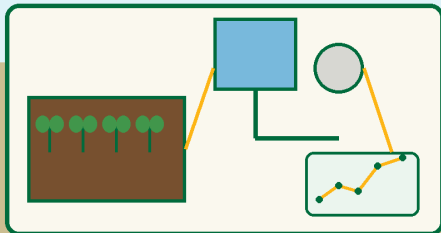
ANALYSIS / DESIGN TASK

Build a conceptual model from the laboratory system.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

SCENARIO ANALYSIS

SCENARIO



SCENARIO

SCENARIO ANALYSIS

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

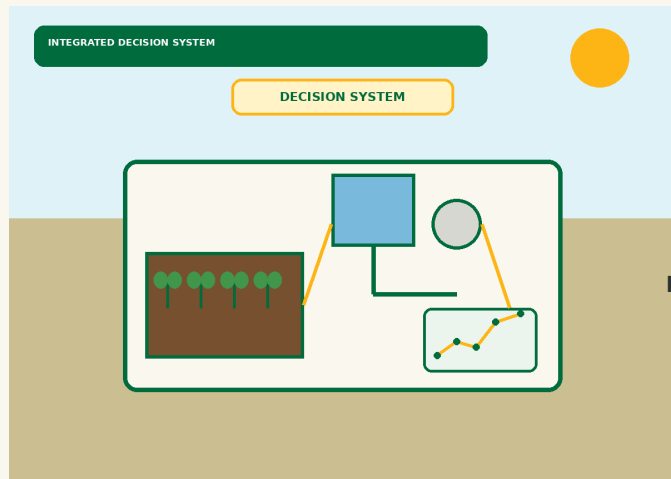
TECHNICAL CONCEPT

Scenario analysis evaluates system performance under alternative resource, demand, failure, or management conditions.

ANALYSIS / DESIGN TASK

Compare baseline and constrained scenarios.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



DECISION SYSTEM

INTEGRATED DECISION SYSTEM

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

An integrated decision system combines measurements, models, constraints, objectives, and operating rules to support coordinated resource management.

ANALYSIS / DESIGN TASK

Propose a decision strategy and evaluate it using multiple performance indicators.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE